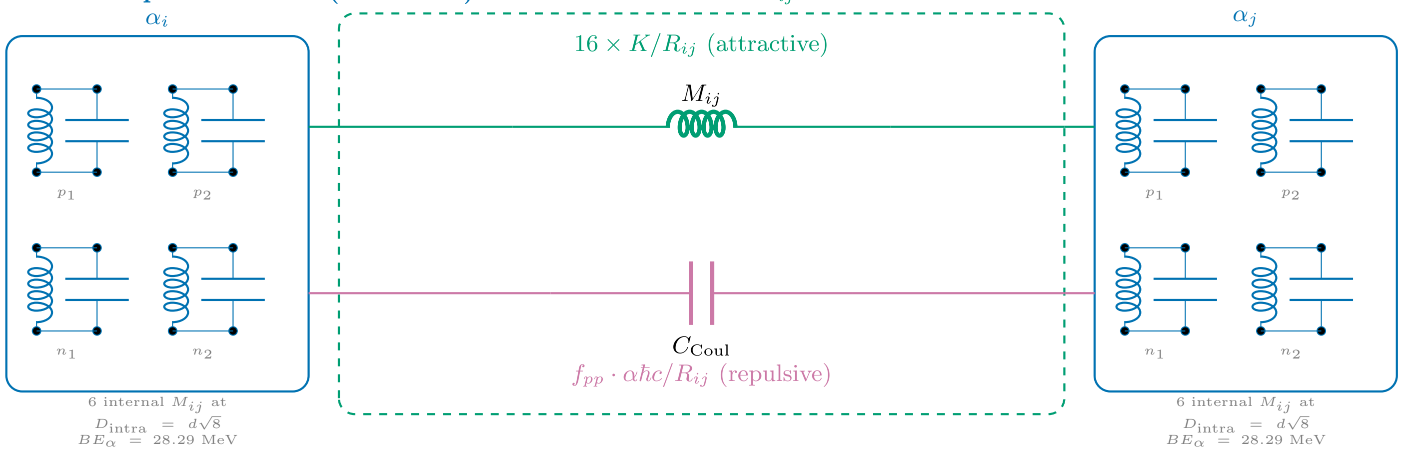
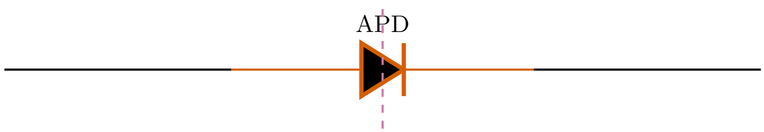
Ti-48 Semiconductor Equivalent Circuit

12α Cuboctahedron — $V_R/V_{BR} = 0.030$ — $M = 1.000$ (Small Signal)

A: Inter-Alpha Junction (Unit Cell)



B: Miller Avalanche Stage

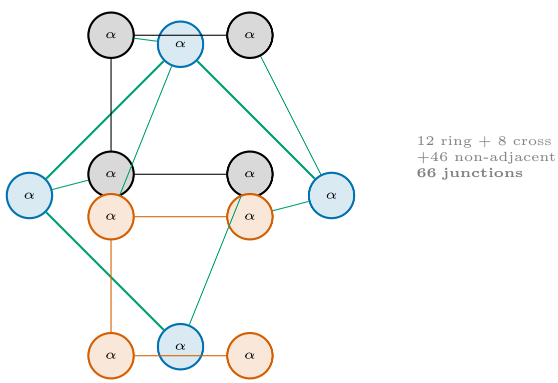


$V_{BR} = 6\alpha\hbar c/D_{\text{intra}} = 3.594$ MeV

$$\boxed{M = \frac{1}{1 - \left(\frac{V_R}{V_{BR}}\right)^5}}$$

Ti-48: $V_R/V_{BR} = 0.030 \implies M = 1.0$ (Deep Small Signal)

C: Ti-48 Topology (66 junctions)



D: Energy Balance

$M_{\text{nuc}} = 12\, M_\alpha - BE_{\text{net}}$

$$BE_{\text{net}} = \underbrace{\sum_{i<j}^{66} 16 \cdot \frac{K}{R_{ij}}}_{textStrong(attractive)} - \underbrace{M \cdot \sum f_{pp} \frac{\alpha\hbar c}{R_{ij}}}_{\text{Coulomb (repulsive)}}$$

Strong: $\sum K/R =$ engine-derived
Coulomb: $\sum \alpha\hbar c/R \times f_{pp}$
Miller: $M = 1.000$ ($V_R/V_{BR} = 0.030$)
Result: $44,636.570$ MeV (0.0001% error)

Each $\alpha = 2p + 2n$ LC tanks at tetrahedron vertices ($D_{\text{intra}} = d\sqrt{8} \approx 2.38$ fm).
$d = 4\hbar/m_p c \approx 0.841$ fm. $K_{\text{mutual}} = 11.34$ MeV·fm. $V_{BR} = 3.594$ MeV.